

## TANUSHREE NEPAL

Missouri, USA

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### EDUCATION

#### **Master of Science in Data Science and Artificial Intelligence**

**12/2026(Ongoing)**

*University of Central Missouri, USA*

*GPA: 3.87 / 4.00*

- **Relevant Coursework:** Artificial Intelligence, Neural Networks and Deep Learning, Machine Learning, Advanced Algorithms, Statistical Foundations for Data Science, Programming Foundations for Data Science

**Thesis:** What Hurts Retrieval-Augmented Generation in Biomedical Question Answering: A Stress-Test Study of Noise, Conflict, and Unanswerability

#### **Master in Business Administration (MBA)**

**08/2025**

*Asia e University, Malaysia*

*GPA: 3.65 / 4.00*

- **Relevant Coursework:** Entrepreneurship and Innovation, Supply Chain Management, Economics, Accounting and Finance for Managers, Enterprise Risk Management, Risk Analysis and Management, Quality and Change Management, Cyber Law
- **Thesis:** Impact of Generative AI on Software Development Students' Creativity and Learning in Kathmandu Valley

#### **Bachelor of Engineering (B.E.) in Computer Engineering**

**09/2022**

*Lalitpur Engineering College, Tribhuvan University, Nepal*

*GPA: 3.64 / 4.00*

- **Relevant Coursework:** Distributed Systems, Database Management Systems, Probability & Statistics, Information Systems, Computer Organization & Architecture, Big Data Technologies, Software Engineering, Computer Networks, Computer Graphics, Numerical Methods, Microprocessors, Data Structures & Algorithms, Object-Oriented Programming
- **Senior Research Project:** Online Cyberbullying Detection Using Natural Language Processing

### PUBLICATIONS

#### **Accepted Papers**

Dadala, K., Nepal, T., Tsai, I Hua

*CodeBERT at the Watering Hole for Dual-Layer URL and JavaScript Attack Detection.*

Accepted at **IEEE EIT 2026**(in proceeding), Main Track. Presented On: May 22, 2026.

**Nepal, T., Shrestha, Y., Ansari, K., Dadala, K., Tsai, I Hua, & Feng, A.**

*Stress-Testing Biomedical Retrieval-Augmented Generation: A Diagnostic Evaluation under Noise, Conflicting Evidence, and Unanswerability.*

Accepted at 2026 **IEEE International Conference on E-health Networking, Application & Services (HealthCom 2026)**, Main Track. Will Present On: Oct 19, 2026

#### **Conference Presentations**

**Nepal, T., Tsai, I Hua**

*What Hurts RAG in Biomedical QA: A Stress-Test Study of Noise, Conflict, and Unanswerability.*

## **Under Review**

**Nepal, T.**, Pamulapati, V. K., Gai, R., Tsai, I Hua, & Feng, A.

Generative AI for Short-Video Feed Recommendation on KuaiRec: A Proxy Study for In-Feed Ad Optimization.

Under review at **IEEE International Conference on Big Data 2026**, Main Track

**Nepal, T.**

Impact of Generative AI on Software Development Students' Creativity and Learning in Kathmandu Valley.

Under review at **Nepalese Journal of Management and Technology**.

## **RESEARCH AND PROJECT EXPERIENCE**

### **WHAT HURTS RETRIEVAL-AUGMENTED GENERATION IN BIOMEDICAL QUESTION ANSWERING: STRESS-TEST STUDY OF NOISE, CONFLICT AND UNANSWERABILITY**

*Master's Thesis under Dr. I Hus Tsai, University of Central Missouri*

*01/2026*

- Built a Retrieval-Augmented Generation (RAG) system for biomedical question answering using the BioASQ dataset, integrating dense retrieval, re-ranking, and LLaMA-3-based answer generation to improve evidence-grounded responses and reduce hallucinations.
- Conducted controlled experiments on noisy, conflicting, and unanswerable queries to analyze failure modes, evaluating system performance using metrics such as MRR, Recall@k, F1, and ROUGE.

**Key skills:** Retrieval-Augmented Generation (RAG), Biomedical NLP, Information Retrieval, Large Language Models, Ranking Metrics (MRR, NDCG), Experimental Design, Statistical Analysis, PyTorch, HuggingFace

### **ALGORITHMIC TRADING & SYNTHETIC INDEX MODELING (KALSHI MARKETS)**

*Independent Research under Dr. Gorkem Kar, University of Central Missouri*

*08/2025*

- Built a high-frequency trading system in Rust for identifying and executing short-term arbitrage opportunities in crypto prediction markets, incorporating a synthetic index model and Deep Reinforcement Learning (DQN, PyTorch) for improved trade timing and probability estimation.
- Performed backtesting and risk-return analysis to evaluate strategy performance and stability, and deployed a scalable, fault-tolerant system with real-time monitoring and a Streamlit dashboard for multi-factor analytics.

**Key skills:** Deep Reinforcement Learning, Quantitative Finance, Market Microstructure, Probabilistic Modeling, Rust Systems Engineering, PyTorch, ONNX Deployment, Statistical Backtesting

### **IMPACT OF GENERATIVE AI ON SOFTWARE DEVELOPMENT STUDENTS' CREATIVITY AND LEARNING IN KATHMANDU VALLEY**

*Independent Researcher*

*01/2025*

- Conducted a quantitative study on the impact of generative AI tools on student creativity, learning outcomes, and engagement among software development students, analyzing relationships between AI familiarity, usage, and academic performance.
- Applied statistical methods including correlation, regression, and mediation analysis (Hayes Model 4), finding a small but significant positive effect on creativity, with no significant impact on learning outcomes or engagement.

## **PROFESSIONAL EXPERIENCE**

### **Graduate Research Assistant**

*University of Central Missouri | Lee's Summit, MO*

*01/2025 — Present*

- Designed and evaluated Retrieval-Augmented Generation (RAG) pipelines for BioASQ biomedical QA, integrating BM25, dense embeddings, reranking, and evidence-constrained answer generation across factoid, list, and yes/no tasks.
- Conducted systematic ablation studies on chunking strategies, embedding models, prompting, and context selection, developing reproducible workflows and evaluating performance using MRR, NDCG, EM, and F1 to ensure retrieval-grounded faithfulness and relevance.

### **Financial Data Analyst**

*Genese Solution Pvt. Ltd. | Nepal*

*12/2022 — 11/2024*

- Built an LLM-driven automation system using Anthropic APIs and AWS CloudFormation to convert natural language prompts into deployable infrastructure templates, improving provisioning efficiency by 20%.
- Developed and validated credit risk forecasting and financial NLP models using Scikit-learn, TensorFlow, and HuggingFace Transformers, improving predictive accuracy by 15% and reducing deployment defects by 90% through standardized validation workflows.

### **Computer Science Teacher**

*Ideal Model School | Nepal*

*03/2023 - 10/2024*

- Designed and delivered C programming and SQL instruction to 60+ students, emphasizing algorithmic thinking and applied problem-solving through hands-on labs and project-based coursework.
- Managed and supervised three computer laboratories, ensuring reliable technical infrastructure and supporting effective programming instruction and experimentation.

## **TECHNICAL SKILLS**

**Programming & Systems:** Python, R, SQL, Git

**Machine Learning & AI:** Deep Learning, Reinforcement Learning (DQN), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Recommendation Systems, Generative AI

**NLP & Information Retrieval:** Hugging Face Transformers, BM25, Cross-Encoder Reranking, Prompt Engineering

**Data Science & Quantitative Methods:** Statistical Modeling, Regression Analysis, Mediation Analysis (Hayes Process Model 4), Time Series Analysis (VAR), Risk Modeling, Credit Risk Forecasting

**Cloud & Data Engineering:** AWS (S3, Glue, Athena, Lambda, Redshift, SageMaker), CloudFormation, Snowflake

**Frameworks & Tooling:** PyTorch, TensorFlow, Scikit-learn, ONNX, LangChain, Streamlit, FastAPI, Pydantic

## **CERTIFICATIONS**

- [AWS Certified AI Practitioner](#)
- [General Registered Engineer](#) **07/2024-Present**

*General Registered Engineer following the Nepal Engineering Council Act (1999 AD) | Registration number: 85011*